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ML OPS Experimentation Stage

Running ML Flow Server and Artifacts URI

```
root@ebb3a0e43a2a:~# mlflow server --backend-store-uri='sqlite:///Assignment/02_training_pipeline/notebooks/lead_scoring_model_experimentation.db'
--default-artifact-root='file:///Assignment/02_training_pipeline/notebooks/mlruns' --port=6006 --host=0.0.0.0
[2025-02-13 17:58:03 +0000] [6024] [INFO] Starting gunicorn 20.1.0
[2025-02-13 17:58:03 +0000] [6024] [INFO] Listening at: http://0.0.0.0:6006 (6024)
[2025-02-13 17:58:03 +0000] [6024] [INFO] Using worker: sync
[2025-02-13 17:58:03 +0000] [6026] [INFO] Booting worker with pid: 6026
[2025-02-13 17:58:03 +0000] [6027] [INFO] Booting worker with pid: 6027
[2025-02-13 17:58:03 +0000] [6028] [INFO] Booting worker with pid: 6028
[2025-02-13 17:58:03 +0000] [6029] [INFO] Booting worker with pid: 6029
```

Model -1: TK_Lead_Scoring_Model_Experimentation

ebb3a02660781.notebooks.jarvislabs.net/#/experiments/1

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Experiments

Search Experiments

Default

TK_Lead_Scoring_Mo...  

TK_Lead_Scoring_Mo...  

Experiment ID: 1

Track machine learning training runs in experiments. [Learn more](#)

Description [Edit](#)

 Refresh  Compare  Delete  Download CSV  AUC  All time 

 Columns Only show differences   metrics.rmse < 1 and params.model = "tree" Search  Filter Clear

Showing 11 matching runs

	Start Time	Duration	Run Name	User	Source	Version	Models	Metrics	Accuracy	F1	Parameters
☐	24 minutes ago		Light Gradient Boosting Machine	root	 ipykerne...	-	 sklearn	0.821	0.738	0.762	-
☐	33 minutes ago		Light Gradient Boosting Machine	root	 ipykerne...	-	 sklearn	0.821	0.738	0.762	-
☐	33 minutes ago		Random Forest Classifier	root	 ipykerne...	-	 sklearn	0.819	0.737	0.76	-
☐	33 minutes ago		Extra Trees Classifier	root	 ipykerne...	-	 sklearn	0.818	0.737	0.758	-
☐	33 minutes ago		Decision Tree Classifier	root	 ipykerne...	-	 sklearn	0.817	0.736	0.758	-
☐	33 minutes ago		Logistic Regression	root	 ipykerne...	-	 sklearn	0.792	0.717	0.741	1.0
☐	33 minutes ago		Linear Discriminant Analysis	root	 ipykerne...	-	 sklearn	0.79	0.715	0.742	-
☐	33 minutes ago		Naive Bayes	root	 ipykerne...	-	 sklearn	0.738	0.679	0.728	-
☐	33 minutes ago		Ridge Classifier	root	 ipykerne...	-	 sklearn	0	0.715	0.742	-
☐	33 minutes ago		Extreme Gradient Boosting	root	 ipykerne...	-	-	-	-	-	-
☐	35 minutes ago		Session Initialized 6c37	root	 ipykerne...	-	-	-	-	-	-

[Load more](#)

ebb3a02660781.notebooks.jarvislabs.net/#/experiments/1/runs/edb384dbf24b4821a951c8d5c8369807

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TK_Lead_Scoring_Model_Experimentation > Light Gradient Boosting Machine

Light Gradient Boosting Machine

Date: 2025-02-13 23:40:00 Source:  ipykernel_launcher.py User: root

Status: UNFINISHED Lifecycle Stage: active Parent Run: 562aaa73fa564c9aaebaf497ace6457c

Description [Edit](#)

Parameters (20)

Metrics (8)

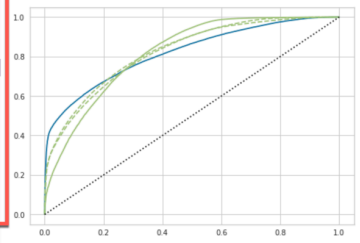
Tags (5)

Artifacts

- model
 - ML_model
 - conda.yaml
 - model.pkl
 - python_env.yaml
 - requirements.txt
 - AUC.png
 - Confusion Matrix.png
 - Feature Importance.png
 - Holdout.html

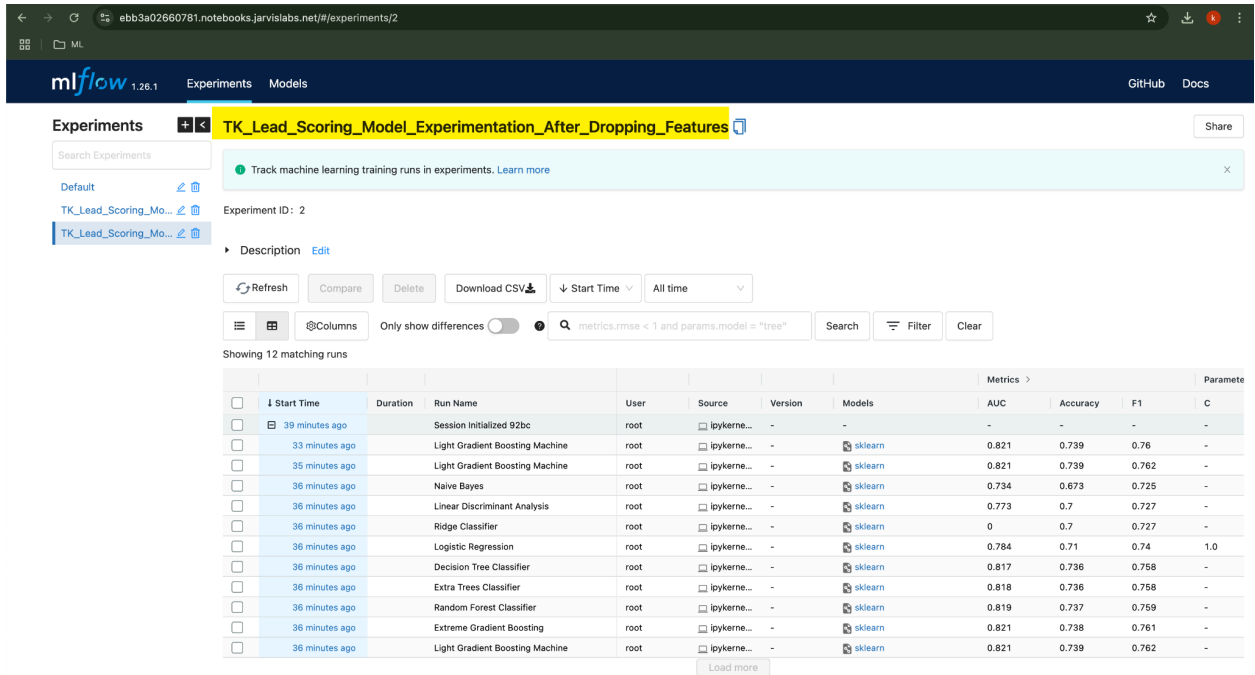
Full Path: file:///Assignment/02_training_pipeline/notebooks/mlruns/1/edb384dbf24b4821a951c8d5c8369807/artifacts/AUC.png 

Size: 27.41KB



Model - 2:

TK_Lead_Scoring_Model_Experimentation_After_Dropping_Features



The screenshot displays the mlflow Experiments interface for the experiment titled "TK_Lead_Scoring_Model_Experimentation_After_Dropping_Features". The experiment ID is 2. The interface shows a list of 12 matching runs, sorted by start time. The runs include a session initialization and various machine learning models trained using sklearn. The metrics for each run are displayed, including AUC, Accuracy, F1, and C.

	↓ Start Time	Duration	Run Name	User	Source	Version	Models	Metrics >	Parameter		
								AUC	Accuracy	F1	C
☐	39 minutes ago		Session Initialized 92bc	root	ipykerne...	-	-	-	-	-	-
☐	33 minutes ago		Light Gradient Boosting Machine	root	ipykerne...	-	sklearn	0.821	0.739	0.76	-
☐	35 minutes ago		Light Gradient Boosting Machine	root	ipykerne...	-	sklearn	0.821	0.739	0.762	-
☐	36 minutes ago		Naive Bayes	root	ipykerne...	-	sklearn	0.734	0.673	0.725	-
☐	36 minutes ago		Linear Discriminant Analysis	root	ipykerne...	-	sklearn	0.773	0.7	0.727	-
☐	36 minutes ago		Ridge Classifier	root	ipykerne...	-	sklearn	0	0.7	0.727	-
☐	36 minutes ago		Logistic Regression	root	ipykerne...	-	sklearn	0.784	0.71	0.74	1.0
☐	36 minutes ago		Decision Tree Classifier	root	ipykerne...	-	sklearn	0.817	0.736	0.758	-
☐	36 minutes ago		Extra Trees Classifier	root	ipykerne...	-	sklearn	0.818	0.736	0.758	-
☐	36 minutes ago		Random Forest Classifier	root	ipykerne...	-	sklearn	0.819	0.737	0.759	-
☐	36 minutes ago		Extreme Gradient Boosting	root	ipykerne...	-	sklearn	0.821	0.738	0.761	-
☐	36 minutes ago		Light Gradient Boosting Machine	root	ipykerne...	-	sklearn	0.821	0.739	0.762	-

Load more

← → ↺ ebb3a02660781.notebooks.jarvislabs.net/#/experiments/2/runs/75761c876d0548c7be88d2bcd8d06e52 ☆ 🔍

ML

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TK_Lead_Scoring_Model_Experimentation_After_Dropping_Features > Light Gradient Boosting Machine

Light Gradient Boosting Machine

Date: 2025-02-13 23:46:05

Source:  ipykernel_launcher.py

User: root

Status: UNFINISHED

Lifecycle Stage: active

Parent Run: 658a893e1e7041deb47ed87daf3c989c

▶ Description [Edit](#)

▶ Parameters (23)

▶ Metrics (8)

▶ Tags (5)

▼ Artifacts

model

 MLmodel

 conda.yaml

 model.pkl

 python_env.yaml

 requirements.txt

 AUC.png

 Confusion Matrix.png

 Feature Importance.png

 Holdout.html

 Iterations.html

Full Path: file:///Assignment/02_training_pipeline/notebooks/miruns/2/75761c876d0548c7be88d2bcd8d06e52/artifacts/model 🔗

Register Model

MLflow Model

The code snippets below demonstrate how to make predictions using the logged model. You can also [register it to the model registry](#) to version control

Model schema

Input and output schema for your model. [Learn more](#)

Name	Type
No schema. See MLflow docs for how to include input and output schema with your model.	

Make Predictions

Predict on a Spark DataFrame:

```
import mlflow
logged_model = 'runs:/75761c876d0548c7be88d2bcd8d06e52/model'

# Load model as a Spark UDF. Override result_type if the model does not return double values.
loaded_model = mlflow.pyfunc.spark_udf(spark, model_uri=logged_model, result_type='double')

# Predict on a Spark DataFrame.
columns = list(df.columns)
df.withColumn('predictions', loaded_model(*columns)).collect()
```